

BILKENT UNIVERSITY

Engineering Mathematics I			
Midterm			
Code : Math 241		Last Name: <i>KE</i>	
Acad. Year: 2021-2022		Name : <i>KE</i>	
Semester : Fall		Student No.: <i>KE</i>	
Date : 22.11.2021		Signature: <i>KE</i>	
Time : 18:00		4 QUESTIONS ON 4 PAGES	
Duration : 100 minutes		TOTAL 100 POINTS	
1	2	3	4

No partial credit will be given for unsupported answers!

1. (13+12 pts) Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$

(a) Find the inverse of A .

(b) Solve $Ax = c$ using Cramer's rule if $c = \begin{bmatrix} 2 \\ 1 \\ 7 \end{bmatrix}$.

(a) $\left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 2 & 5 & 3 & 0 & 1 & 0 \\ 1 & 0 & 8 & 0 & 0 & 1 \end{array} \right] \xrightarrow{\substack{R_2 \rightarrow -2R_1 + R_2 \\ R_3 \rightarrow -R_1 + R_3}} \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 1 & 0 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & -2 & 5 & -1 & 0 & 1 \end{array} \right]$

$\begin{array}{l} R_1 \rightarrow -2R_2 + R_1 \\ R_3 \rightarrow 2R_2 + R_3 \end{array} \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 9 & 5 & -2 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & -1 & -5 & 2 & 1 \end{array} \right] \xrightarrow{R_3 \rightarrow -R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 9 & 5 & -2 & 0 \\ 0 & 1 & -3 & -2 & 1 & 0 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right]$

$\begin{array}{l} R_1 \rightarrow -9R_3 + R_1 \\ R_2 \rightarrow 3R_3 + R_2 \end{array} \rightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -40 & 16 & 9 \\ 0 & 1 & 0 & 13 & -5 & -3 \\ 0 & 0 & 1 & 5 & -2 & -1 \end{array} \right]$

$\underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}}_{I_3} \quad \underbrace{\begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}}_{A^{-1}}$

$A^{-1} = \begin{bmatrix} -40 & 16 & 9 \\ 13 & -5 & -3 \\ 5 & -2 & -1 \end{bmatrix}$

b)

$$x_1 = \frac{\begin{vmatrix} 2 & 2 & 3 \\ 1 & 5 & 3 \\ 7 & 0 & 8 \end{vmatrix}}{\begin{vmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{vmatrix}} = \frac{7 \begin{vmatrix} 2 & 3 \\ 5 & 3 \end{vmatrix} + 8 \begin{vmatrix} 2 & 2 \\ 1 & 5 \end{vmatrix}}{1 \begin{vmatrix} 2 & 3 \\ 5 & 3 \end{vmatrix} + 8 \begin{vmatrix} 1 & 2 \\ 2 & 5 \end{vmatrix}} = \frac{-63 + 64}{-9 + 8} = -1$$

$$x_2 = \frac{\begin{vmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \\ 1 & 7 & 8 \end{vmatrix}}{\begin{vmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{vmatrix}} = \frac{1 \begin{vmatrix} 1 & 3 \\ 7 & 8 \end{vmatrix} - 2 \begin{vmatrix} 2 & 3 \\ 1 & 8 \end{vmatrix} + 3 \begin{vmatrix} 2 & 1 \\ 1 & 7 \end{vmatrix}}{-1} = 0$$

$$x_3 = \frac{\begin{vmatrix} 1 & 2 & 2 \\ 2 & 5 & 1 \\ 1 & 0 & 7 \end{vmatrix}}{\begin{vmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{vmatrix}} = \frac{1 \begin{vmatrix} 2 & 2 \\ 5 & 1 \end{vmatrix} + 7 \begin{vmatrix} 1 & 2 \\ 2 & 5 \end{vmatrix}}{-1} = \frac{-8 + 7}{-1} = 1$$

2. (10+10+5 pts) Let

$$A = \begin{bmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & -1 & 1 \\ 1 & 2 & -1 & 4 \end{bmatrix}$$

- (a) Find a basis for $\text{Col}(A)$.
 (b) Find a basis for $\text{Null}(A)$.
 (c) Find rank and nullity of A .

$$\left[\begin{array}{cccc|c} 1 & 0 & 1 & 2 & 0 \\ 0 & 1 & -1 & 1 & 0 \\ 1 & 2 & -1 & 4 & 0 \end{array} \right] \xrightarrow{R_1 \rightarrow -R_1 + R_2} \left[\begin{array}{cccc|c} 1 & 0 & 1 & 2 & 0 \\ 0 & 1 & -1 & 1 & 0 \\ 0 & 2 & -2 & 2 & 0 \end{array} \right] \xrightarrow{R_3 \rightarrow -2R_2 + R_3}$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 1 & 2 & 0 \\ 0 & 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right] \quad \begin{array}{l} \text{let } x_1, x_2, x_3, x_4 \text{ be the variables.} \\ \text{then } x_1, x_2 \text{ leading var.} \\ x_3 = \alpha, x_4 = \beta \text{ free variables.} \end{array}$$

$$(a) \text{Col}(A) = \text{Span} \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 4 \end{bmatrix} \right\}$$

$$(b) \text{ If } x_3 = \alpha, x_4 = \beta \quad \text{then} \quad \begin{array}{l} x_2 = \alpha - \beta \\ x_1 = -\alpha - 2\beta \end{array} \Rightarrow \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{bmatrix} = \begin{bmatrix} -\alpha - 2\beta \\ \alpha - \beta \\ \alpha \\ \beta \end{bmatrix}$$

$$\Rightarrow \text{Null}(A) = \left\{ \begin{bmatrix} -\alpha - 2\beta \\ \alpha - \beta \\ \alpha \\ \beta \end{bmatrix} : \forall \alpha, \beta \in \mathbb{R} \right\}$$

$$= \text{Span} \left\{ \begin{bmatrix} -1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ -1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$$(c) \text{rank}(A) = \dim(\text{Col}(A)) = 2$$

$$\text{nullity}(A) = \dim(\text{Null}(A)) = 2$$

3. (25 pts) Let $A = [a_1 \ a_2 \ a_3]$ be a 3×3 matrix where $a_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, $a_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$. Find a column vector a_3 of length 1 that will maximize $\det A$.

Since $|\det A| = \text{volume of the parallelepiped whose sides are } a_1, a_2, a_3$

This volume will be maximized if a_3 is orthogonal to $\text{Span}\{a_1, a_2\}$. If $a_3 = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, $a_3 \cdot a_1 = 0$
 $a_3 \cdot a_2 = 0$

$$\Rightarrow \begin{cases} x+y+z=0 \\ x-y+z=0 \end{cases} \Rightarrow \begin{cases} y=0 \\ z=x \end{cases}, x=-\alpha \text{ where } \alpha \in \mathbb{R}.$$

$$\Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \alpha \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \quad \text{For } \alpha = \frac{1}{\sqrt{2}}, \text{ let } a_3 = \begin{bmatrix} -1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix}$$

Then $\|a_3\| = 1$ and $|a_1 \ a_2 \ a_3|$ will be maximum

4. (10+10+5 pts) Let P_3 be the set of polynomials of degree less than or equal to 3 with real coefficients. Define

$$W_1 = \{p(x) \in P_3 \mid p(1) = 0\}$$

$$W_2 = \{p(x) \in P_3 \mid p(0) = 0\}$$

- (a) Show W_1 is a subspace of P_3 .
 (b) Find a basis for $W_1 \cap W_2$.
 (c) Find $\dim(W_1 + W_2)$.

(a) • Since $p(x) = 0 \in W_1$, $W_1 \neq \emptyset$.

• Let $p_1(x), p_2(x) \in P_3 \Rightarrow p_1(1) = 0$
 $p_2(1) = 0$.

Then $(p_1 + p_2)(1) = p_1(1) + p_2(1) = 0 \Rightarrow (p_1 + p_2)(x) \in P_3$.

• Let $p_1(x) \in P_3 \Rightarrow p_1(1) = 0$.

Then $(\alpha p_1)(1) = \alpha p_1(1) = 0$.

Since $W_1 \neq \emptyset$, it is closed under addition and scalar multiplication, W_1 is a subspace of P_3 .

(b) Let $p(x) = ax^3 + bx^2 + cx + d$ be an arbitrary element of P_3 . If $p(x) \in W_1 \cap W_2$ then $p(1) = a + b + c + d = 0$
 $p(0) = d = 0$.

$$\begin{aligned} \Rightarrow W_1 \cap W_2 &= \{ax^3 + bx^2 + cx \mid a + b + c = 0; a, b, c \in \mathbb{R}\} \\ &= \{ax^3 + bx^2 + (-a-b)x \mid a, b \in \mathbb{R}\} \\ &= \{a(x^3 - x) + b(x^2 - x) \mid a, b \in \mathbb{R}\} \\ &= \text{span}\{x^3 - x, x^2 - x\} \end{aligned}$$

Since $\alpha(x^3 - x) + \beta(x^2 - x) = 0 \Rightarrow \alpha = \beta = 0$,

$\{x^3-x, x^2-x\}$ is a basis for $\omega_1 \cap \omega_2$.

$$(c) \dim(\omega_1 + \omega_2) \leq \min \left\{ \underbrace{\dim(P_3)}_{\substack{\text{"} \\ 4}}, \underbrace{\dim \omega_1 + \dim \omega_2}_{\substack{3 \\ \leftarrow \text{new problem} \\ \text{set 3}}} \right\}$$

$$\Rightarrow \text{Since } \left. \begin{array}{l} \omega_1 \neq \omega_1 + \omega_2 \\ \omega_2 \neq \omega_1 + \omega_2 \end{array} \right\} \left\{ \underbrace{\dim \omega_1}_{\substack{\text{"} \\ 3}}, \underbrace{\dim \omega_2}_{\substack{\text{"} \\ 3}} < \dim(\omega_1 + \omega_2) \right.$$

$$\Rightarrow 3 < \dim(\omega_1 + \omega_2) \leq 4$$

$$\Rightarrow \dim(\omega_1 + \omega_2) = 4$$